

STAGE 1: INPUT & PERCEPTION

Food Image

- Overhead plate meal
- Multi-dish meal scene

Food Detection

YOLOv8n Food Detector

- 123 visual food classes
- 56,147 annotated images

Packaged Food

- Commercial food item
- UPC / EAN / GTIN barcode

Barcode Decoding

Barcode Decoder

- Product identity lookup
- Open Food Facts & local DB

Nutrition Label

- Menu description
- Packaged facts panel

Text Extraction

OCR Engine

- Text line token parsing
- Serving size & nutrient regex

STAGE 2: CANONICAL REPRESENTATION & KNOWLEDGE GROUNDING

Canonical Identity

Unified Entity

- Name normalization
- Synonym/alias map
- Portion/unit scale
- Confidence fusion

Nutrition Grounding

Clinical Database

- Calories & macros
- Protein, Carbs, Fats
- Micronutrients & fiber
- IFCT / USDA database

Regional Knowledge

Grounding (217 Classes)

- 217 regional classes
- 10 culinary categories
- Recipes & ingredients
- Dietary & cultural tags

STAGE 3: CONSTRAINED OPTIMIZATION & OUTPUT

Constrained Meal Planning

OR-Tools / SCIP Solver

- 4 Meal Slots: B / L / D / Snack
- Calorie & macro bounds: $(1 \pm \tau)T$
- Dietary & ingredient constraints
- Candidate selection: $N \in [50, 500]$

Nutrition-Grounded Meal Plan

Optimized Feasible Output

- Optimized daily recipe schedule
- Target nutrient & calorie fulfillment
- Serving sizes & portion weights
- Nutritional summary & compliance